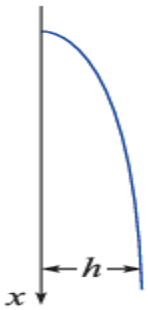
Name: Enrolment No:	UPES UNIVERSITY OF TOMORROW

UPES

Assignment-2

Programme Name : B. Tech. SoCS/SoAE **Semester : II**
Course Name : Advanced Engineering Mathematics-II **Max. Marks: 100**
Course Code : MATH1065 **Unit(s) : III-V**

Date of Submission: 30/04/2026

S. No.		Marks	CO
1	Verify the function $u = aexp\left[-\left(\frac{x}{h} - \frac{ct}{h}\right)^2\right]$ satisfies the wave equation $\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}$. Sketch the graph of the solution u against x at $t = 0, t = {}^{2h}/_{c}$ and , $t = 4h\ /\ c$.	**15**	**CO5**
2	A thin layer of liquid drains down the side of a vessel, as illustrated in the figure below. From the theory of thin layers, the equation for the fluid motion is given by $$\frac{\partial h}{\partial t} + ah^2\frac{\partial h}{\partial x} = 0$$ where $h(x,t)$ is the thickness of the layer and a is the constant that depends on the viscosity, density and the gravity constant. Find the solution for $h(x,t)$ given the initial condition $h(x,0) = \alpha\sqrt{x}$ where α is a constant. 	**15**	**CO5**
3	Solve the linear partial differential equation $$\frac{\partial^2 z}{\partial x^2} + 2\frac{\partial^2 z}{\partial x \partial y} + \frac{\partial^2 z}{\partial y^2} = \sin(2x + 3y).$$	**10**	**CO5**
4	The equation of motion of the moving coil of a galvanometer when a current i is passed through it is of the form $$\frac{d^2\theta}{dt^2} + 2K\frac{d\theta}{dt} + n^2\theta = \frac{n^2 i}{K}$$ where θ is the angle of deflection from the 'no-current' position and n and K are positive constants. Given that i is a constant and $\theta = 0 = d\theta/dt$ when $t = 0$, obtain an expression for the Laplace transform of θ. In constructing the galvanometer, it is desirable to have it critically damped (that is, $n = K$).	**15**	**CO4**

	Use the Laplace transform method to solve the differential equation in this case and sketch the graph of θ against t for positive values of t.		
5	A periodic function $f(t)$ of period 2π is defined within the period $0 \le t \le 2\pi$ by $$f(t) = \begin{cases} t & 0 \le t \le \frac{\pi}{2} \\ \frac{\pi}{2} & \frac{\pi}{2} \le t \le \pi \\ \pi - \frac{t}{2} & \frac{\pi}{2} \le t \le 2\pi \end{cases}$$ Sketch the graph of the function $f(t)$ for $-2\pi \le t \le 3\pi$ and find a Fourier series of it in $0 \le t \le 2\pi$.	**15**	**CO4**
6	A pellet of mass m is fired from a gun at time $t = 0$ with a muzzle velocity v_0. If the pellet is fired into a viscous gas, the equation of motion can be expressed as $$m\frac{d^2x}{dt^2} + k\frac{dx}{dt} = mv_0\delta(t); x(0) = x'(0) = 0,$$ Where $x(t)$ is the displacement at time $t \ge 0$, and k is constant. Find the displacement using Laplace transform.	**10**	**CO4**
7	Consider the differential equation $$(x-1)(x-2)(x-3)(x-4)(x-5)^2y'' + (x^2-1)y' + y = 0$$ Find and classify all the singular points of the equation.	**10**	**CO3**
8	Discuss what are Bessel functions $J_0(x)$ and $J_1(x)$. Also, find Laplace Transform of these Bessel functions.	**10**	**CO3**